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054323 - INTERNET OF THINGS

POLITECNICO DI MILANO

COMPUTER SCIENCE AND ENGINEERING

IoT Challenge 4 - Homework
Exercise 3

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1 Exercise 3

1.1 System Description and Given Data

A RFID system employs **Dynamic Frame ALOHA** to resolve $N = 4$ tags. In each frame of size r, every tag independently selects one of the r slots uniformly at random and transmits its ID in that slot. Slots are classified as:

- **Empty**: no tag transmits → wasted slot.
- **Success**: exactly one tag transmits → tag identified and removed from the backlog.
- **Collision**: two or more tags transmit → all colliding tags remain in the backlog.

After the first frame with size r_1, the frame size is set exactly to the current backlog size for all subsequent frames. The **collision resolution efficiency** is defined as:

$$\eta = \frac{N}{L_{\text{total}}} \tag{1}$$

where L_{total} is the expected total number of slots required to identify all N tags. The following values are given:

Given	Value
L_1	1
L_2	4
L_3	$51/8 = 6.375$

Table 1: Given arbitration period durations ($r = N$)

where L_k denotes the expected total number of slots to resolve k tags when the initial frame size is set optimally to $r = k$. $L_1 = 1$ trivially (one tag, one slot, always a success).

1.2 Derivation of L_4

To compute L_4 (the expected duration to resolve $N = 4$ tags with optimal frame $r = 4$), we first derive the backlog distribution after the first frame.

1.2.1 Backlog Distribution for $N = 4$, $r = 4$

With 4 tags and 4 slots there are $4^4 = 256$ equally probable configurations. We enumerate all possible slot occupation patterns (Table 2).

Pattern	**Successes**	**Backlog** b	**Configurations**	$P(b)$
$(1,1,1,1)$	4	0	$4! = 24$	24/256
$(2,1,1,0)$	2	2	$\binom{4}{1}\binom{4}{2}P(3,2) = 4 \cdot 6 \cdot 6 = 144$	144/256
$(3,1,0,0)$	1	3	$\binom{4}{1}\binom{4}{3} \cdot 3 = 4 \cdot 4 \cdot 3 = 48$	48/256
$(2,2,0,0)$	0	4	$\binom{4}{2}\binom{4}{2} = 6 \cdot 6 = 36$	36/256
$(4,0,0,0)$	0	4	4	4/256
Total			**256**	**1**

Table 2: Slot occupation patterns for $N = 4$, $r = 4$

Aggregating over backlog values:

$$P(b = 0 \mid N = 4, r = 4) = \frac{24}{256} = \frac{3}{32} \tag{2}$$

$$P(b = 2 \mid N = 4, r = 4) = \frac{144}{256} = \frac{9}{16} \tag{3}$$

$$P(b = 3 \mid N = 4, r = 4) = \frac{48}{256} = \frac{3}{16} \tag{4}$$

$$P(b = 4 \mid N = 4, r = 4) = \frac{36 + 4}{256} = \frac{5}{32} \tag{5}$$

1.2.2 Recursive Formula for L_4

The total expected duration satisfies the recursion ($L_0 = 0$):

$$L_4 = 4 + P(b = 0) \cdot L_0 + P(b = 2) \cdot L_2 + P(b = 3) \cdot L_3 + P(b = 4) \cdot L_4 \tag{6}$$

Substituting:

$$L_4 = 4 + \frac{3}{32} \cdot 0 + \frac{9}{16} \cdot 4 + \frac{3}{16} \cdot \frac{51}{8} + \frac{5}{32} \cdot L_4 \tag{7}$$

$$L_4 = 4 + \frac{9}{4} + \frac{153}{128} + \frac{5}{32} L_4 = 7.4453 + \frac{5}{32} L_4 \tag{8}$$

Solving:

$$L_4\left(1 - \frac{5}{32}\right) = 7.4453 \quad \Rightarrow \quad L_4 \cdot \frac{27}{32} = 7.4453 \quad \Rightarrow \quad \boxed{L_4 = \frac{7.4453 \times 32}{27} \approx 8.824} \tag{9}$$

1.3 Efficiency for Each Initial Frame Size r_1

For each $r_1 \in \{1, 2, 3, 4, 5, 6\}$ we compute the backlog distribution after the first frame ($N = 4$ tags, r_1 slots, r_1^4 equiprobable configurations), then apply:

$$L_{\text{total}}(r_1) = r_1 + \sum_b P(b \mid N = 4, r_1) \cdot L_b \qquad \eta(r_1) = \frac{4}{L_{\text{total}}(r_1)} \tag{10}$$

1.3.1 Case $r_1 = 1$

With a single slot all 4 tags must collide; the backlog is always 4.

$$P(b = 4) = 1 \tag{11}$$

$$L_{\text{total}} = 1 + L_4 = 1 + 8.824 = 9.824 \qquad \eta = \frac{4}{9.824} \approx \mathbf{0.4073} \tag{12}$$

1.3.2 Case $r_1 = 2$

With $r_1 = 2$ and $N = 4$ there are $2^4 = 16$ configurations. Each tag independently picks slot 1 or 2 with probability $\frac{1}{2}$. The number of tags in slot 1 follows $\text{Bin}(4, \frac{1}{2})$. A slot is a success only if it contains exactly one tag.

k in slot 1	$4-k$ in slot 2	Successes	b	Configs
0	4	0	4	1
1	3	1	3	4
2	2	0	4	6
3	1	1	3	4
4	0	0	4	1

Table 3: Backlog distribution for $r_1 = 2$

$$P(b = 3) = \frac{8}{16} = \frac{1}{2}, \qquad P(b = 4) = \frac{8}{16} = \frac{1}{2} \tag{13}$$

$$L_{\text{total}} = 2 + \frac{1}{2} \cdot \frac{51}{8} + \frac{1}{2} \cdot 8.824 = 2 + 3.1875 + 4.412 = 9.600 \qquad \eta = \frac{4}{9.600} \approx \mathbf{0.4167} \tag{14}$$

1.3.3 Case $r_1 = 3$

With $r_1 = 3$ and $N = 4$ there are $3^4 = 81$ configurations.

Pattern	**Successes**	b	**Configurations**	$P(b)$
$(4,0,0)$	0	4	3	3/81
$(3,1,0)$	1	3	$3 \cdot 2 \cdot \binom{4}{3} = 24$	24/81
$(2,2,0)$	0	4	$3 \cdot \binom{4}{2} = 18$	18/81
$(2,1,1)$	2	2	$3 \cdot \binom{4}{2} \cdot 2! = 36$	36/81
Total			**81**	**1**

Table 4: Backlog distribution for $r_1 = 3$

$$P(b=2) = \frac{36}{81} = \frac{4}{9}, \quad P(b=3) = \frac{24}{81} = \frac{8}{27}, \quad P(b=4) = \frac{21}{81} = \frac{7}{27} \tag{15}$$

$$\begin{aligned} L_{\text{total}} &= 3 + \frac{4}{9} \cdot 4 + \frac{8}{27} \cdot \frac{51}{8} + \frac{7}{27} \cdot 8.824 \\ &= 3 + 1.7778 + 1.8889 + 2.2878 = 8.954 \qquad \eta = \frac{4}{8.954} \approx \mathbf{0.4467} \end{aligned} \tag{16}$$

1.3.4 Case $r_1 = 4$

This is the case where $r_1 = N = 4$, already fully analysed in Section 2. Using $L_4 = 8.824$:

$$L_{\text{total}} = L_4 = 8.824 \qquad \eta = \frac{4}{8.824} \approx \mathbf{0.4533} \tag{17}$$

1.3.5 Case $r_1 = 5$

With $r_1 = 5$ and $N = 4$ there are $5^4 = 625$ configurations. Since $r_1 > N$, the pattern $(1,1,1,1,0)$, all four tags in distinct slots with one slot empty, now has non-zero probability, meaning full resolution in the first frame is possible.

Pattern	b	**Configurations**	$P(b)$
$(4,0,0,0,0)$	4	5	5/625
$(3,1,0,0,0)$	3	$5 \cdot 4 \cdot \binom{4}{3} = 80$	80/625
$(2,2,0,0,0)$	4	$\binom{5}{2} \cdot \binom{4}{2} = 60$	60/625
$(2,1,1,0,0)$	2	$5 \cdot \binom{4}{2} \cdot \binom{4}{2} \cdot 2! = 360$	360/625
$(1,1,1,1,0)$	0	$5 \cdot 4! = 120$	120/625
Total		**625**	**1**

Table 5: Backlog distribution for $r_1 = 5$

$$P(b=0) = \frac{24}{125}, \quad P(b=2) = \frac{72}{125}, \quad P(b=3) = \frac{16}{125}, \quad P(b=4) = \frac{13}{125} \tag{18}$$

$$\begin{aligned} L_{\text{total}} &= 5 + \frac{24}{125} \cdot 0 + \frac{72}{125} \cdot 4 + \frac{16}{125} \cdot \frac{51}{8} + \frac{13}{125} \cdot 8.824 \\ &= 5 + 2.304 + 0.816 + 0.9177 = 9.038 \qquad \eta = \frac{4}{9.038} \approx \mathbf{0.4426} \end{aligned} \tag{19}$$

1.3.6 Case $r_1 = 6$

With $r_1 = 6$ and $N = 4$ there are $6^4 = 1296$ configurations.

Pattern	b	**Configurations**	$P(b)$
$(4,0,0,0,0,0)$	4	6	6/1296
$(3,1,0,0,0,0)$	3	$6 \cdot 5 \cdot \binom{4}{3} = 120$	120/1296
$(2,2,0,0,0,0)$	4	$\binom{6}{2} \cdot \binom{4}{2} = 90$	90/1296
$(2,1,1,0,0,0)$	2	$6 \cdot \binom{5}{2} \cdot \binom{4}{2} \cdot 2! = 720$	720/1296
$(1,1,1,1,0,0)$	0	$\binom{6}{2} \cdot 4! = 360$	360/1296
Total		**1296**	**1**

Table 6: Backlog distribution for $r_1 = 6$

$$P(b=0) = \frac{5}{18}, \quad P(b=2) = \frac{5}{9}, \quad P(b=3) = \frac{5}{54}, \quad P(b=4) = \frac{2}{27} \tag{20}$$

$$\begin{aligned} L_{\text{total}} &= 6 + \frac{5}{18} \cdot 0 + \frac{5}{9} \cdot 4 + \frac{5}{54} \cdot \frac{51}{8} + \frac{2}{27} \cdot 8.824 \\ &= 6 + 2.2222 + 0.5903 + 0.6536 = 9.466 \qquad \eta = \frac{4}{9.466} \approx \mathbf{0.4227} \end{aligned} \tag{21}$$

1.4 Summary Table and Plot

r_1	L_{total}	η	Notes
1	9.824	0.4073	All tags collide: worst case
2	9.600	0.4167	
3	8.954	0.4467	
4	**8.824**	**0.4533**	$r_1 = N$: **maximum**
5	9.038	0.4426	One empty slot on average
6	9.466	0.4227	Too many wasted slots

Table 7: Collision resolution efficiency η as a function of r_1

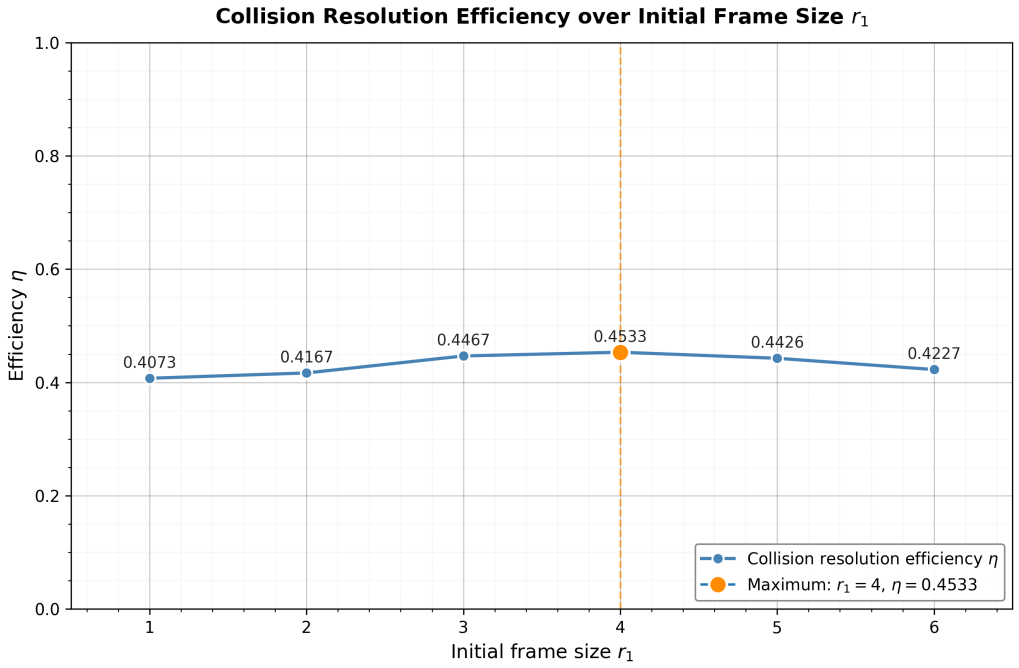


Figure 1: Collision resolution efficiency η vs. initial frame size r_1 for $N = 4$ tags

1.5 Maximum Efficiency and Discussion

Result - Maximum Efficiency

The maximum efficiency is achieved at $r_1 = 4 = N$:

$$\eta_{\max} = \frac{4}{8.824} \approx \mathbf{0.4533}$$

1.5.1 Why is $r_1 = N$ Optimal?

The result $r_1 = N$ is consistent with the well-known theoretical optimum for Frame ALOHA, which can be understood intuitively as follows.

- **When** $r_1 < N$: there are more tags than slots, so the expected number of collisions per frame is high. Many tags are forced to share the same slot, creating large residual backlogs that require many additional frames to resolve. The extra frames add heavily to L_{total}, reducing efficiency.
- **When** $r_1 = N$: each slot "expects" exactly one tag on average. This balances the trade-off between collisions (too few slots) and wasted empty slots (too many slots). The probability of a success in any given slot is maximised at this point, yielding the highest fraction of useful slot usage.
- **When** $r_1 > N$: the frame is unnecessarily large. Many slots remain empty (no tag chose them) and no useful transmission occurs in those slots. Since empty slots contribute to L_{total} but resolve zero tags, efficiency drops. Furthermore, the probability that all 4 tags fall in distinct slots increases, so the residual backlog after the first frame is often zero, but the first frame alone already "cost" more slots than necessary.

In summary, setting $r_1 = N$ minimises L_{total} because it maximises the **expected number of successful slots per frame**, which is exactly the throughput of the Frame ALOHA channel. The efficiency curve in Figure 1 confirms this: η increases monotonically from $r_1 = 1$ to $r_1 = 4$, then decreases for $r_1 > 4$, with a clear peak at $r_1 = N = 4$.